

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Part – VII – Karnaugh Map(K-map) Simplification of an Boolean expression

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Karnaugh - Maps (K-Map) Representation

Video Link - <https://youtu.be/B2ivIMueC2E>

- During the process of simplification of Boolean expression we have to predict each successive steps
- We can never be absolutely certain that an expression simplified by Boolean algebra alone is the simplest possible expression
- On the other hand, the map method gives us a systematic approach for simplifying a Boolean expression
- The map method, first proposed by Veitch and modified by Karnaugh, hence it is known as the Veitch diagram or the Karnaugh map.

One Variable Maps (2 cells)

	A	
Value of A	0	
	1	

Two Variables Maps

A \ B	0	1
0		
1		

Three Variable Maps

A \ BC	00	01	11	10
0				
1				

Four Variable Maps

AB \ CD	00	01	11	10
00				
01				
11				
10				

Values of Gray Code Sequence

1, 2, 3 and 4 Variables Maps with Product terms

1 Variable Map

$\bar{A}$	$\bar{A}_0$
A	A_1

2-Variable Map

	A	B	$\bar{B}$	B
$\bar{A}$	$\bar{A}\bar{B}_0$	$\bar{A}B_1$		
A	$A\bar{B}_2$	AB_3		

3 Variable Map

	A	BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	$\bar{A}\bar{B}\bar{C}_0$	$\bar{A}\bar{B}C_1$	$\bar{A}B\bar{C}_3$	$\bar{A}BC_2$		
A	$A\bar{B}\bar{C}_4$	$A\bar{B}C_5$	$AB\bar{C}_7$	ABC_6		

4-Variable Map

	AB	CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	$\bar{A}\bar{B}\bar{C}\bar{D}_0$	$\bar{A}\bar{B}\bar{C}D_1$	$\bar{A}\bar{B}C\bar{D}_3$	$\bar{A}\bar{B}CD_2$		
$\bar{A}B$	$\bar{A}B\bar{C}\bar{D}_4$	$\bar{A}B\bar{C}D_5$	$\bar{A}BC\bar{D}_7$	$\bar{A}BCD_6$		
AB	$AB\bar{C}\bar{D}_{12}$	$AB\bar{C}D_{13}$	$ABC\bar{D}_{15}$	$ABCD_{14}$		
$A\bar{B}$	$A\bar{B}\bar{C}\bar{D}_8$	$A\bar{B}\bar{C}D_9$	$A\bar{B}C\bar{D}_{11}$	$A\bar{B}CD_{10}$		

1, 2, 3, and 4 Variable maps with Sum terms

0 A	A 0
1 $\bar{A}$	$\bar{A}$ 1

1-Variable Map

	B	$\bar{B}$
A	A+B 0	A+ $\bar{B}$ 1
$\bar{A}$	$\bar{A}$ +B 2	$\bar{A}$ + $\bar{B}$ 3

2-Variable Map

A \ BC	B + C	B + $\bar{C}$	$\bar{B}$ + $\bar{C}$	$\bar{B}$ + C
$\bar{A}$	A + B + C 0	A + B + $\bar{C}$ 1	A + $\bar{B}$ + $\bar{C}$ 3	A + $\bar{B}$ + C 2
$\bar{A}$	$\bar{A}$ + B + C 4	$\bar{A}$ + B + $\bar{C}$ 5	$\bar{A}$ + $\bar{B}$ + $\bar{C}$ 7	$\bar{A}$ + $\bar{B}$ + C 6

3-Variable Map

AB \ CD	C + D	C + $\bar{D}$	$\bar{C}$ + $\bar{D}$	$\bar{C}$ + D
A + B	A + B + C + D 0	A + B + C + $\bar{D}$ 1	A + B + $\bar{C}$ + $\bar{D}$ 3	A + B + $\bar{C}$ + D 2
A + $\bar{B}$	A + $\bar{B}$ + C + D 4	A + $\bar{B}$ + C + $\bar{D}$ 5	A + $\bar{B}$ + $\bar{C}$ + $\bar{D}$ 7	A + $\bar{B}$ + $\bar{C}$ + D 6
$\bar{A}$ + $\bar{B}$	$\bar{A}$ + $\bar{B}$ + C + D 12	$\bar{A}$ + $\bar{B}$ + C + $\bar{D}$ 13	$\bar{A}$ + $\bar{B}$ + $\bar{C}$ + $\bar{D}$ 15	$\bar{A}$ + $\bar{B}$ + $\bar{C}$ + D 14
$\bar{A}$ + B	$\bar{A}$ + B + C + D 8	$\bar{A}$ + B + C + $\bar{D}$ 9	$\bar{A}$ + B + $\bar{C}$ + $\bar{D}$ 11	$\bar{A}$ + B + $\bar{C}$ + D 10

4-Variable Map

1, 2, 3 and 4 Variable maps for SOP expressions

- minterms

		B	
		0	1
A	0	m_0 0	m_1 1
	1	m_2 2	m_3 3

1-Variable Map

2-Variable Map

		BC			
		00	01	11	10
A	0	m_0 0	m_1 1	m_3 3	m_2 2
	1	m_4 4	m_5 5	m_7 7	m_6 6

3-Variable Map

		CD			
		00	01	11	10
AB	00	m_0 0	m_1 1	m_3 3	m_2 2
	01	m_4 4	m_5 5	m_7 7	m_6 6
	11	m_{12} 12	m_{13} 13	m_{15} 15	m_{14} 14
	10	m_8 8	m_9 9	m_{11} 11	m_{10} 10

4-Variable Map

1, 2, 3 and 4 Variable maps for POS Expressions - Maxterms

A	0	M ₀ 0
	1	M ₁ 1

1 - Variable Map

		B	
		0	1
A	0	M ₀ 0	M ₁ 1
	1	M ₂ 2	M ₃ 3

2 - Variable Map

		BC			
		00	01	11	10
A	0	M ₀ 0	M ₁ 1	M ₃ 3	M ₂ 2
	1	M ₄ 4	M ₅ 5	M ₇ 7	M ₆ 6

3 - Variable Map

		CD			
		00	01	11	10
AB	00	M ₀ 0	M ₁ 1	M ₃ 3	M ₂ 2
	01	M ₄ 4	M ₅ 5	M ₇ 7	M ₆ 6
	11	M ₁₂ 12	M ₁₃ 13	M ₁₅ 15	M ₁₄ 14
	10	M ₈ 8	M ₉ 9	M ₁₁ 11	M ₁₀ 10

4 - Variable Map

Plotting a Karnaugh Map

Video Link - <https://youtu.be/QYP5kgA95uc>

Logic function can be represented in various forms such as truth table, SOP boolean expression and POS boolean expression. In this section we will see the procedures to plot the given logic function in any form on the Karnaugh map.

Representation of Truth Table on Karnaugh Map

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

A \ B	0	1
0	0	1
1	1	0

A \ B	$\bar{B}$	B
$\bar{A}$	0	1
A	1	0

Representation of Standard SOP on K-map

A boolean expression in the sum of products form can be plotted on the Karnaugh map by placing a 1 in each cell corresponding to a term (minterm) in the sum of products expression. Remaining cells are filled with zeros.

Example #01

Plot Boolean Expression

$Y = AB\bar{C} + ABC + \bar{A}\bar{B}C$ on the Karnaugh Map

A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	0	0
A	0	0	1	1

Example #02

Plot Boolean expression on the Karnaugh Map

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}C\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D}$$

AB \ CD				
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	0	0	0
$\bar{A}B$	1	0	0	1
AB	0	1	0	0
$A\bar{B}$	0	0	1	1

Representation of Standard Pos on k-map

A boolean expression in the product of sums can be plotted on the Karnaugh map by placing a 0 in each cell corresponding to a term (maxterm) in the expression. Remaining cells are filled with ones.

Example #01

Plot boolean expression $Y = (A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)$

$(A + B + \bar{C})$ on the Karnaugh map

A \ BC				
	$B+C$	$B+\bar{C}$	$\bar{B}+\bar{C}$	$\bar{B}+C$
A	1	0	0	0
$\bar{A}$	1	1	1	0

Example #02

Plot the boolean expression

$$Y = (A + B + C + \bar{D})(A + \bar{B} + \bar{C} + D)(A + B + \bar{C} + \bar{D})(\bar{A} + \bar{B} + \bar{C} + D)(\bar{A} + \bar{B} + C + \bar{D})$$

AB \ CD				
	$C+D$	$C+\bar{D}$	$\bar{C}+\bar{D}$	$\bar{C}+D$
$A+B$	1	0	0	1
$A+\bar{B}$	1	1	1	0
$\bar{A}+\bar{B}$	1	0	1	0
$\bar{A}+B$	1	1	1	

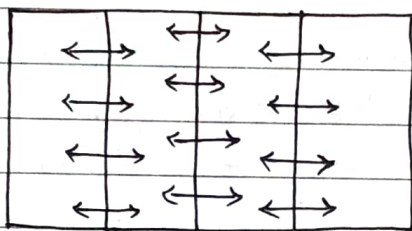
Grouping Cells for Simplification

Video Link - <https://youtu.be/G860zPuftL4>

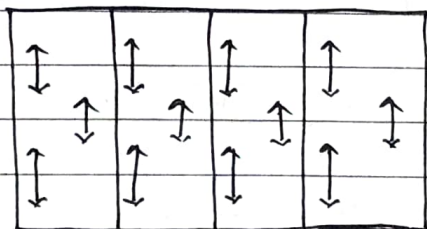
→ Once the Boolean function is plotted on the Karnaugh map we have to use grouping technique to simplify the boolean function

→ The grouping is nothing but combining terms in adjacent cells

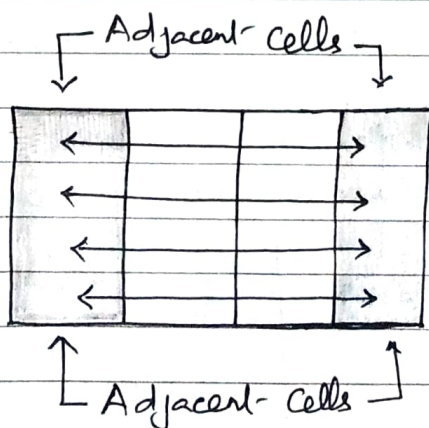
(a) Neighboring cells in the row are adjacent



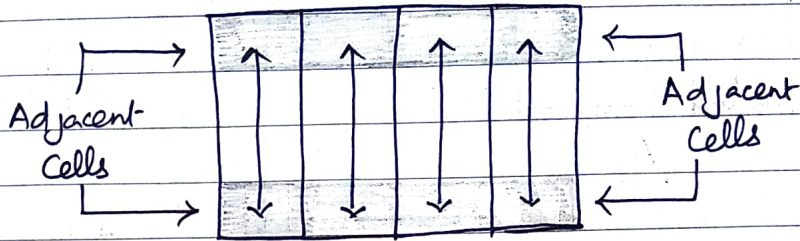
(b) Neighbouring cells in the Column are adjacent



(c) Leftmost and Corresponding rightmost cells are adjacent



(d) Top and Bottom Corresponding Bottom Cells are adjacent



(i) Grouping Two Adjacent Ones (Pair)

A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	0	0

$\bar{A}C$

A \ BC	00	01	11	10
0	0	0	1	0
1	0	0	1	0

BC

A \ BC	00	01	11	10
0	0	0	0	0
1	1	0	0	1

$A\bar{C}$

A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	0

$\bar{A}C + BC$

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	0	0	0
11	0	0	0	0
10	0	1	0	0

$\bar{B}\bar{C}D$

A \ BC	00	01	11	10
0	0	1	1	0
1	0	0	1	1

Group 1 (circled 1s at 01, 11 in row 0)
Group 2 (circled 1s at 11, 10 in row 1)
Pair not required

$\bar{A}C + AB$

(ii) Grouping Four Adjacent Ones (Quad)

A \ BC	00	01	11	10
	0	0	0	0
0	0	0	0	0
1	1	1	1	1

A

AB \ CD	00	01	11	10
	0	0	1	0
00	0	0	1	0
01	0	0	1	0
11	0	0	1	0
10	0	0	1	0

CD

AB \ CD	00	01	11	10
	0	0	0	0
00	0	0	0	0
01	0	1	1	0
11	0	1	1	0
10	0	0	0	0

BD

AB \ CD	00	01	11	10
	0	0	0	0
00	0	0	0	0
01	0	0	0	0
11	1	0	0	1
10	1	0	0	1

$\overline{A}D$

AB \ CD	00	01	11	10
	0	0	0	0
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

$\overline{B}\overline{D}$

AB \ CD	00	01	11	10
	0	0	0	0
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

Group 1 Group 2 Group 3

$AB + \overline{A}D + AC$

(iii) Grouping Eight Adjacent Ones (octet)

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

B

AB \ CD	CD			
	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

D

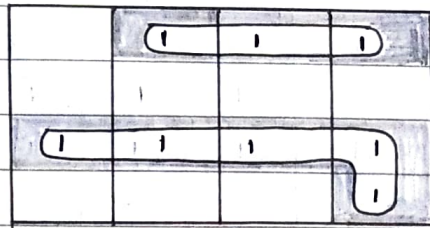
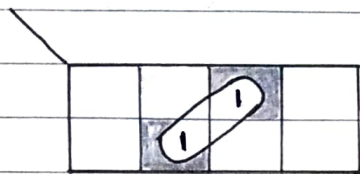
AB \ CD	CD			
	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

$\overline{B}$

AB \ CD	CD			
	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	0	1
10	1	0	0	1

$\overline{D}$

(iv) Illegal Grouping



Simplification SOP function Using K-maps

Video Link - <https://youtu.be/dqKbT5uNrIM>

- A pair of 1's eliminates one variable, a quad of 1s eliminates two variables and an octet of 1s eliminates three variables
- In general, when a variable appears in both complemented and uncomplemented form within a group, that variable is eliminated from the resultant expression
- Variables that are same in all within the group must appear in the final expression
- Each group gives us a product term and summation of all product terms gives us a Boolean expression
- Each product term implies the function and, hence is an implicant of the function
- All the implicants of a function determined using a Karnaugh map are the prime implicants

From the above discussion we can outline generalized procedure to simplify Boolean expression as follows.

- (i) Plot the K-map and place 1s in those cells corresponding to the 1s in the truth table or sum of expressions. Place 0s in other cells
- (ii) Check the K-map for adjacent 1s and encircle those 1s which are not adjacent to any other 1s. These are called isolated 1s
- (iii) Check for those 1's which are adjacent to only one 1 and encircle such pairs
- (iv) check for quads and octets of adjacent 1s even if it contains some 1s that have already been encircled. While doing this make sure that there are minimum number of groups
- (v) Combine any pairs necessary to include any 1s that have not yet been grouped
- (vi) Form the simplified expression by summing product terms of all the groups

Example #01

Minimize the expression by using k-map

$$Y = A\bar{B}C + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

Step 1:

Diagram shows the k-map for three variables and it is plotted according to the given expression

A \ BC				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	1 ₀	1 ₁	1 ₃	0 ₂
A	1 ₄	1 ₅	0 ₇	0 ₆

$$Y = \bar{B} + \bar{A}C$$

Step 2: There are no isolated 1s

Step 3: 1 in cell 3 is adjacent only to 1 in the cell 1.
This pair is combined and referred to as group 1

Step 4: There is no octet, but there is a quad.
Cells 0, 1, 4 and 5 form a quad. This quad is combined and referred to as group 2

Step 5: All 1s have already been grouped

Step 6: Each group generates a term in the expression for Y. In group 1, B variable is eliminated.
and in group 2, variables A and C are eliminated and we get,
$$Y = \bar{A}C + B$$

Example #02Video Link - <https://youtu.be/1vrRNJira1A>

Minimize the expression by using K-map

$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D}$$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	0	0	1
$\bar{A}B$	1	1	0	0
AB	1	1	0	0
$A\bar{B}$	0	1	0	0

$$Y = \bar{A}\bar{B}C\bar{D} + B\bar{C} + A\bar{C}D$$

Example #03Video Link - <https://youtu.be/uISZec6ZpuY>

Reduce the following four variable function to its minimum sum of products form

$$Y = \bar{A}\bar{B}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D}$$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	0	1	1
$\bar{A}B$	0	0	0	0
AB	1	0	0	1
$A\bar{B}$	1	0	1	1

$$Y = \bar{B}\bar{D} + A\bar{D} + \bar{B}C$$

Example #04

Video Link - <https://youtu.be/v1imNMABLXE>

Reduce the following function to its minimum sum of product form

$$Y = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}BCD + AB\bar{C}\bar{D} + AB\bar{C}D + ABCD + A\bar{B}CD$$

AB \ CD				
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	0	0
$\bar{A}B$	0	1	1	1
AB	1	1	1	0
$A\bar{B}$	0	0	1	0

$$Y = \bar{A}\bar{C}D + \bar{A}BC + AB\bar{C} + ACD$$

Example #05

Video Link - <https://youtu.be/mhrL7V02PMA>

Simplify the logic function specified by the truth table, using the Karnaugh map method.

Y is the output Variable, A, B and C are the input Variables

A	B	C	Y	
0	0	0	1	0
0	0	1	0	1
0	1	0	0	2
0	1	1	1	3
1	0	0	1	4
1	0	1	0	5
1	1	0	0	6
1	1	1	1	7

A \ BC		00	01	11	10
		$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
0 $\overline{A}$	1	0	1	0	
1 A	1	0	1	0	

$$Y = \bar{B}\bar{C} + BC$$

Example #06

Video Link - <https://youtu.be/9wtDsukpxmU>

Reduce the following function using Karnaugh map technique and implement using basic gates

$$f(A, B, C, D) = \bar{A}\bar{B}D + AB\bar{C}\bar{D} + \bar{A}BD + ABC\bar{D}$$

The given function is not the standard sum of Product form. It is converted into standard SOP form as follows

$$\begin{aligned} f(A, B, C, D) &= \bar{A}\bar{B}D + AB\bar{C}\bar{D} + \bar{A}BD + ABC\bar{D} \\ &= \bar{A}\bar{B}D(C + \bar{C}) + AB\bar{C}\bar{D} + \bar{A}BD(C + \bar{C}) + ABC\bar{D} \\ &= \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}D + AB\bar{C}\bar{D} + \bar{A}BCD + \bar{A}B\bar{C}D + ABC\bar{D} \end{aligned}$$

AB \ CD		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	1	0	
$\bar{A}B$	0	1	1	0	
AB	1	0	0	1	
$A\bar{B}$	0	0	0	0	

$$f = \bar{A}D + AB\bar{D}$$

Example #07

Video Link - <https://youtu.be/hJfXrxgbMkA>

Reduce the following function using K-map techniques

$$f(A, B, C, D) = \sum m(0, 1, 4, 8, 9, 10)$$

AB \ CD	00		01		11		10	
	$\bar{C}\bar{D}$		$\bar{C}D$		CD		$C\bar{D}$	
00 $\bar{A}\bar{B}$	1	1	0	1	0	3	0	2
01 $\bar{A}B$	1	0	0	5	0	7	0	6
11 AB	0	0	0	13	0	15	0	14
10 $A\bar{B}$	1	1	0	9	0	11	1	10

$$f(X) = \bar{A}\bar{C}\bar{D} + A\bar{B}\bar{D} + \bar{B}\bar{C}$$

Example #08

Video Link - https://youtu.be/IPYPsj_Jq58

Obtain the simplified expression in Sum of Product for the following boolean functions

$$F = \sum m(0, 1, 4, 5, 10, 11, 12, 14)$$

AB \ CD	00		01		11		10	
	$\bar{C}\bar{D}$		$\bar{C}D$		CD		$C\bar{D}$	
00 $\bar{A}\bar{B}$	1	1	0	1	0	3	0	2
01 $\bar{A}B$	1	1	0	5	0	7	0	6
11 AB	1	0	0	13	0	15	1	14
10 $A\bar{B}$	0	0	1	9	1	11	1	10

$$f(X) = \bar{A}\bar{C} + A\bar{B}\bar{D} + A\bar{B}C$$

Example #09

Video Link - <https://youtu.be/yRuy8R8bkP0>

Obtain the simplified expression in Sum of Product for the following boolean function

$$F = \sum m(11, 12, 13, 14, 15)$$

AB \ CD				
	00 $\bar{C}\bar{D}$	01 $\bar{C}D$	11 CD	10 $C\bar{D}$
00 $\bar{A}\bar{B}$	0 0	0 1	0 3	0 2
01 $\bar{A}B$	0 4	0 5	0 7	0 6
11 AB	1 12	1 13	1 15	1 14
10 $A\bar{B}$	0 8	0 9	1 11	0 10

$$f(x) = AB + ACD$$

Example #10

Video Link - <https://youtu.be/Pv6oAao5Bw>

Simplify the boolean function with Karnaugh map

$$F(w, x, y, z) = \sum (0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

wx \ yz				
	00 $\bar{y}\bar{z}$	01 $\bar{y}z$	11 yz	10 $y\bar{z}$
00 $\bar{w}\bar{x}$	1 0	1 1	0 3	1 2
01 $\bar{w}x$	1 4	1 5	0 7	1 6
11 wx	1 12	1 13	0 15	1 14
10 $w\bar{x}$	1 8	1 9	0 11	0 10

$$f = \bar{y} + \bar{w}\bar{z} + x\bar{z}$$

Example #11

Video Link - <https://youtu.be/VLWjhrmUnVo>

Simplify the boolean function using K-map

$$F = \bar{A}\bar{B}\bar{C} + \bar{B}C\bar{D} + \bar{A}Bc\bar{D} + A\bar{B}\bar{C}$$

AB \ CD	CD			
	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$	CD
00 $\bar{A}\bar{B}$	1	1	0	1
01 $\bar{A}B$	0	0	0	1
11 AB	0	0	0	0
10 $A\bar{B}$	1	1	0	1

$$F = \bar{B}\bar{C} + \bar{B}\bar{D} + \bar{A}C\bar{D}$$

Example #12

Video Link - <https://youtu.be/veGe9bDtuuQ>

Reduce to simplest form using K-map

$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10)$$

AB \ CD	CD			
	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$	CD
00 $\bar{A}\bar{B}$	1	1	0	1
01 $\bar{A}B$	0	1	0	0
11 AB	0	0	0	0
10 $A\bar{B}$	1	1	0	1

$$F = \bar{B}\bar{C} + \bar{B}\bar{D} + \bar{A}C\bar{D}$$

Simplification of POS Functions Using K-maps

Video Link - <https://youtu.be/TjZCikPsOxM>

- In practice, the designer should examine both the sum of products and product of sums reductions to ascertain which is more simplified
 - Once the expression is plotted on the k-maps instead of making the groups of ones, we have to make groups of Zeros
 - Each group of Zeros results a sum terms and it is nothing but the prime implicant.
 - The technique for using maps for POS reductions is a simple step by step process and it is similar to the one used earlier
1. Plot the k-map for adjacent and place 0s in those cells corresponding to the 0s in the truth table or Maxterms in the product of Sums expression
 2. Check the k-map for adjacent 0s and encircle those 0s which are not adjacent to any other 0s. These are called isolated 0s
 3. Check for those 0s which are adjacent to only one other 0 and encircle such pairs
 4. Check for quads and octets of adjacent 0s even if it contains some 0s that have already been encircled. while doing this make sure that there are minimum number of groups
 5. Combine any pairs necessary to include any 0s that have not yet been grouped

6. From the Simplified POS expression for F by taking product of sum terms of all the groups

Example # 01

Minimize the expression

$$Y = (A + B + \bar{C})(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + \bar{C})(A + B + C)(\bar{A} + B + C)$$

A \ BC				
	B+C	B+ $\bar{C}$	$\bar{B}$ + $\bar{C}$	$\bar{B}$ +C
A	0	0	0	
$\bar{A}$	0		0	

$$Y = (B + C)(\bar{B} + \bar{C})(A + \bar{C})$$

Example # 02

Video Link - https://youtu.be/2a-Bm_iEAPs

Minimize the following expression in the POS form

$$Y = (\bar{A} + \bar{B} + C + D)(\bar{A} + \bar{B} + \bar{C} + D)(\bar{A} + \bar{B} + \bar{C} + \bar{D})(\bar{A} + B + C + D)(A + \bar{B} + \bar{C} + D)(A + \bar{B} + \bar{C} + \bar{D})(A + B + C + D)(\bar{A} + \bar{B} + C + \bar{D})$$

AB \ CD				
	C+D	C+ $\bar{D}$	$\bar{C}$ + $\bar{D}$	$\bar{C}$ +D
A+B	0	1	3	2
A+ $\bar{B}$	4	5	7	6
$\bar{A}$ + $\bar{B}$	12	13	15	14
$\bar{A}$ +B	8	9	11	10

$$\bar{A} + \bar{B} + C + D = M_{12}$$

$$\bar{A} + \bar{B} + \bar{C} + D = M_{14}$$

$$\bar{A} + \bar{B} + \bar{C} + \bar{D} = M_{15}$$

$$\bar{A} + B + C + D = M_8$$

$$\bar{A} + \bar{B} + \bar{C} + D = M_6$$

$$A + \bar{B} + \bar{C} + \bar{D} = M_7$$

$$Y = (B + C + D)(\bar{A} + \bar{B})(\bar{B} + \bar{C})$$

$$A + B + C + D = M_0$$

$$\bar{A} + \bar{B} + C + \bar{D} = M_{13}$$

Example #03

Video Link - <https://youtu.be/bVbBqyfUHD0>

Reduce the following function using K-maps

$$f(A, B, C, D) = \prod M(0, 2, 3, 8, 9, 12, 13, 15)$$

AB \ CD				
	C+D	C+D̄	C̄+D̄	C̄+D
A+B	0	1	3	2
A+B̄	4	5	7	6
Ā+B̄	12	13	15	14
Ā+B	8	9	11	10

$$F = (\bar{A} + \bar{B} + \bar{D})(A + B + \bar{C})(\bar{A} + C)(A + B + D)$$

Example #04

Video Link - <https://youtu.be/r-PV8HIE62Q>

Reduce the expression in SOP and POS form using K-map

$$F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$$

SOP Form

AB \ CD				
	C̄D̄	C̄D	CD	C̄D̄
ĀB̄		1		X
ĀB	X	1		1
AB	1	1		1
ĀB̄				

$$F = B\bar{C} + B\bar{D} + \bar{A}\bar{C}\bar{D}$$

$$F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$$

$$\therefore F(A, B, C, D) = \prod M(0, 3, 7, 8, 9, 10, 11, 15) + d(2, 4)$$

AB \ CD	C+D	C+D̄	C̄+D̄	C̄+D
A+B	0 ○	1	3 ○	2 X
A+B̄	4 X	5	7 ○	6
Ā+B̄	12	13	15 ○	14
Ā+B	8 ○	9 ○	11 ○	10 ○

$$\underline{F = (\bar{A} + B)(\bar{C} + \bar{D})(B + C + D)}$$

Essential Prime Implicants

Video Link - <https://youtu.be/iFVKFaoTkV0>

- After grouping the cells, the sum terms which appear in the k-map are called prime implicants groups
- It is observed that some cells may appear in only one prime implicants groups, which other cells may appear in more than one prime implicants group.
- In the diagram shown below, cells 1, 4, 9 and 10 appear in only one prime implicants group. These cells are called essential cells and corresponding prime implicants are called essential prime implicants

AB \ CD		00	01	11	10
		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
00	$\bar{A}\bar{B}$	1	1	0	0
01	$\bar{A}B$	1	0	0	0
11	AB	0	0	0	0
10	$A\bar{B}$	1	1	0	1

Diagram illustrating a 4x4 Karnaugh Map (K-map) for a 4-variable function. The map shows cells 1, 4, 9, and 10 as essential cells, each appearing in only one prime implicant group. The groups are highlighted with colored lines: a blue vertical line for cells 1 and 4, a green horizontal line for cells 1 and 9, a red horizontal line for cells 9 and 10, and a red vertical line for cells 10 and 11. The map is labeled with AB and CD, and the cells are numbered 0 through 15.

Incompletely Specified Functions - Don't Care Terms

Video Link - <https://youtu.be/ow6H9mTBP0I>

→ In some logic circuit, certain input conditions never occur, therefore the corresponding output level is not defined, it can be either HIGH or Low. These output levels are indicated by 'x' or 'd' in the truth table and are called don't care outputs or don't care conditions or incompletely specified functions.

	A	B	C	Y
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	x
7	1	1	1	x

→ As shown in truth table, outputs from 000 to 101 are defined. For remaining two conditions of input, output is not defined, hence these are called don't care conditions of this truth table.

→ A circuit designer is free to make the output for any "don't care" condition either 0 or 1 in order to produce the simplest output expression.

$$f(A, B, C) = \sum m(1, 3, 5) + d(6, 7)$$

Example #01

Video Link - <https://youtu.be/2mRgIPeogMk>

Find the reduced SOP form of the following function

$$f(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 4)$$

AB \ CD				
	00 $\bar{C}\bar{D}$	01 $\bar{C}D$	11 CD	10 $C\bar{D}$
00 $\bar{A}\bar{B}$	X 0	1 1	1 3	X 2
01 $\bar{A}B$	X 4	0 5	1 7	0 6
11 AB	0 12	0 13	1 15	0 14
10 $A\bar{B}$	0 8	0 9	1 11	0 10

$$f(A, B, C, D) = \bar{A}\bar{B} + CD$$

Example #02

Video Link - <https://youtu.be/Ffkv9QR5Po0>

Find the prime implicants for the following function and determine which are essential

$$F(w, x, y, z) = \sum (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$$

AB \ CD				
	00 $\bar{C}\bar{D}$	01 $\bar{C}D$	11 CD	10 $C\bar{D}$
00 $\bar{A}\bar{B}$	1 0	1 1	1 3	1 2
01 $\bar{A}B$	4	1 5	1 7	6
11 AB	12	1 13	1 15	14
10 $A\bar{B}$	1 8	9	11	1 10

Essential Prime implicants

$$\bar{x}\bar{z} \text{ and } xz$$

- The diagram shows two essential prime implicants
- The term XZ is essential because there is only one way to include minterms m_{13} and m_{15} within four adjacent squares
- There is only one way that minterms m_8 and m_{10} can be combined with four adjacent squares and this gives second term $\bar{X}\bar{Z}$
- The two essential prime implicants cover eight minterms
- The remaining two minterms m_4 and m_6 must be considered next.

$WX \backslash YZ$				
	$\bar{Y}\bar{Z}$	$\bar{Y}Z$	YZ	$Y\bar{Z}$
$00 \bar{W}\bar{X}$	1 0	1	3	2
$01 \bar{W}X$	4	5	7	6
$11 WX$	12	13	15	14
$10 W\bar{X}$	8	9	11	10

Prime Implicants
 $\bar{W}X$ and $\bar{W}\bar{Z}$

- The above diagram shows the two possible ways of including remaining two minterms with prime implicants
- Minterms m_4 and m_6 can be included with either $\bar{W}\bar{X}$ or $\bar{W}\bar{Z}$

→ The simplified expression is obtained from the logical sum of two essential prime implicants and any one prime implicants that includes minterms m_4 and m_6

→ There are two possible ways that the function can be expressed in the simplified form

$$F = \bar{x}\bar{z} + xz + \bar{w}x = \bar{x}\bar{z} + xz + \bar{w}\bar{z}$$

Example # 03

Video Link - <https://youtu.be/CZ43CDKwk7o>

Express the following function as the minimal sum of products using a K-map

$$f(a, b, c, d) = \sum(0, 2, 4, 5, 6, 8, 10, 15) + \sum\phi(7, 13, 14)$$

Alternative 1

ab \ cd	00	01	11	10
	$\bar{c}\bar{d}$	$\bar{c}d$	cd	$c\bar{d}$
00 $\bar{a}\bar{b}$	1			1
01 $\bar{a}b$	1	1	X	1
11 ab		X	1	X
10 $a\bar{b}$	1			1

$$\underline{f(a, b, c, d) = \bar{a}b + bd + \bar{b}\bar{d}}$$

Alternative 2

ab \ cd	00 $\bar{c}\bar{d}$	01 $\bar{c}d$	11 cd	10 $c\bar{d}$
00 $\bar{a}\bar{b}$	1 0	1	3	2
01 $\bar{a}b$	1 4	1 5	X 7	1 6
11 ab		X 13	1 15	X 14
10 $a\bar{b}$	1 8			1 10

$$\underline{f(a, b, c, d) = \bar{a}b + bc + \bar{b}\bar{d}}$$

Alternative 3

ab \ cd	00 $\bar{c}\bar{d}$	01 $\bar{c}d$	11 cd	10 $c\bar{d}$
00 $\bar{a}\bar{b}$	1 0	1	3	2
01 $\bar{a}b$	1 4	1 5	X 7	1 6
11 ab		X 13	1 15	X 14
10 $a\bar{b}$	1 8			1 10

$$\underline{f(a, b, c, d) = \bar{a}\bar{d} + bd + \bar{b}\bar{d}}$$

Quine McCluskey Method of Function Realization

Video Link - <https://youtu.be/Y1O9-d6TXn0>

1. List all minterms in the binary form
2. Arrange the minterms according to number of 1s
3. Compare each binary number with every term in the adjacent next higher Category and if they differ only by one position, put a check mark and copy the term in the next Column with '-' in the position that they differed
4. Apply the same process described in step 3 for the resultant Column and continue these cycles until a single pass through cycle yields no further eliminations of literals
5. List all prime implicants
6. Select the minimum number of prime implicants which must cover all the minterms

7.

Example 01

Simplify the following Boolean function by using a Quine McCluskey method

$$F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 12, 13)$$

Step 1: List all minterms in binary form as shown as Column (a)

Step 2: Arrange the minterms according to number of 1's as shown in table Column (b)

Column (a)					Column (b)				
Minterms	Binary Representation				Minterms	Binary Representation			
m ₀	0	0	0	0	m ₀	0	0	0	0 ✓
m ₂	0	0	1	0	m ₂	0	0	1	0 ✓
m ₃	0	0	1	1	m ₈	1	0	0	0 ✓
m ₆	0	1	1	0	m ₃	0	0	1	1 ✓
m ₇	0	1	1	1	m ₆	0	1	1	0 ✓
m ₈	1	0	0	0	m ₁₀	1	0	1	0 ✓
m ₁₀	1	0	1	0	m ₁₂	1	1	0	0 ✓
m ₁₂	1	1	0	0	m ₇	0	1	1	1 ✓
m ₁₃	1	1	0	1	m ₁₃	1	1	0	1 ✓

Step 3: Compare each binary number with every term in the adjacent next Higher category and if they differ only by one position, put a check marks and Copy the term in the next Column with '-' in the position that they differed

Step 4: Apply the same process described in Step 3 for the resultant Column and Continue these Cycles until a single pass through Cycle yields no further elimination of literals

Column (c)		Column (d)	
Minterms	Binary Representation	Minterm	Binary Representation
0, 2	0 0 - 0 ✓	0, 2, 8, 10	- 0 - 0
0, 8	- 0 0 0 ✓	2, 3, 6, 7	0 - 1 -
2, 3	0 0 1 - ✓		
2, 6	0 - 1 0 ✓		
2, 10	- 0 1 0 ✓		
8, 10	1 0 - 0 ✓		
8, 12	1 - 0 0		
3, 7	0 - 1 1 ✓		
6, 7	0 1 1 - ✓		
12, 13	1 1 0 -		

Step 5: List the prime implicants

Prime Implicants		Binary Representation
$A \bar{C} \bar{D}$	8, 12	1 - 0 0
$A B \bar{C}$	12, 13	1 1 0 -
$\bar{B} \bar{D}$	0, 2, 8, 10	- 0 - 0
$\bar{A} C$	2, 3, 6, 7	0 - 1 -

Step 6: Select the minimum number of prime implicants which must cover all the minterms.

Table shows prime implicant selection chart. Each prime implicant is represented in a row and each minterm in a ~~row~~ column. Dots are placed in each row to show the composition of minterms that makes the prime implicants. From this chart we have to select the minimum number of prime implicants which must cover all minterms. The selection procedure is as follows.

Search for single dot columns and select the prime implicants corresponding to that dot by putting the check mark in front of it.

Prime Implicants	m ₀ Col 1	m ₂ Col 2	m ₃ Col 3	m ₆ Col 4	m ₇ Col 5	m ₈ Col 6	m ₁₀ Col 7	m ₁₂ Col 8	m ₁₃ Col 9
$A\bar{C}\bar{D}$ 8, 12						•		•	
$AB\bar{C}$ 12, 13 ✓								⊙	⊙
$\bar{B}\bar{D}$ 0, 2, 8, 10 ✓	⊙	⊙				⊙	⊙		
$\bar{A}C$ 2, 3, 6, 7 ✓		⊙	⊙	⊙	⊙				

Search for multiple dot Columns one by one. If the corresponding minterms are already included in the final expression ignore the minterm and go to next multi-dot Column, otherwise include the corresponding prime implicant in the final expression.

→ Column 1 (m₀) has single dot, so include corresponding prime implicant (0, 2, 8, 10)

→ Column 3 (m₃) has single dot so include corresponding prime implicant (2, 3, 6, 7).

→ Columns 4, 5 and 7 have single dots but the corresponding minterms are already included in the final expression

→ Column 9 has single dot so include corresponding prime implicants (12, 13) in the final expression

→ Columns 2, 6 and 8 have multi-dots but their minterms are already included in the final expression, Therefore, the final expression is

$$F(A, B, C, D) = (110-) + (-0-0) + (0-1-)$$

$$F(A, B, C, D) = AB\bar{C} + \bar{B}\bar{D} + \bar{A}C$$